

PROJECT SUTRA — THE GRAND FINALE

COOKBOOK

A First-Principles Operational Recipe Book for 300/300 Marks at Smart Horizon 2026

EVENT & VENUE	TEAM IDENTITY	PROBLEM STATEMENT	TOTAL SCORE POOL
NHCE Library (DST Track)	SHIH26-TID-361	SH-DST-05 (Swarm SAR)	300 Marks (3 × 100)

Chapter 1: The Essential Kitchen Laws (Invariants & Rules)

Recipe 1.1: The "No-Uncommitted-Work" Push Routine CARDINAL LAW

Difficulty: Zero Tolerance **Prep Time:** 15 seconds **Cook Time:** Continuous **Auditor:** Git Remote Master

ACTIVE RULE INVARIANT

"No teammate is allowed to work on local and say 'I didn't commit, but I completed the work.' If work is not committed and pushed to GitHub, it officially does not exist."

- 1 Check Branch:** Verify your active branch before typing: `git branch --show-current`. Work strictly inside your assigned subsystem role branch.
- 2 Test Verification:** Run deterministic tests to confirm zero regressions: `pytest sutra_ws/src/sutra_*/test/ -q`.
- 3 Stage Atomic Diffs:** Never leave uncommitted diffs: `git add <modified_files>`.
- 4 Semantic Commit:** Use standard prefixes: `feat:`, `fix:`, `test:`, `docs:`.
- 5 Immediate Push:** Push immediately: `git push origin <assigned-branch>`. Work is invalid without a live commit hash.

Recipe 1.2: The Zero-Mock Empirical Evidence Filter SCIENTIFIC MOAT

Target: Absolute Defense **Verified Baseline:** 232/232 Tests Passing (15.71s execution)

Strict Policy: Every benchmark in DOCS.md or pitch slides MUST appear verbatim in captured stdout. If an external sensor or drone is offline, write `? UNTESTED - [Reason]`. Never synthesize fake numbers.

Recipe 1.3: 24/7 Desk Attendance & Active Documentation NHCE RULE 3.4 & 6.1

Venue: Library Desk **Min Occupancy:** ≥ 2 Teammates **Active Document:** `JURY_FEEDBACK_TRACKER.md`

- 1 Desk Attendance Invariant:** The table in the Library must NEVER be left empty. Rohith Kumar & Harika anchor the desk during all meal shifts.
- 2 Active Jury Ledger:** During Evaluation 1 & 2, Harika transcribes every question into `JURY_FEEDBACK_TRACKER.md`. Code resolutions must be committed with tags before the subsequent round (Rule 6.1).

Chapter 2: Subsystem Recipes (Preparation & Plating)

Recipe 2.1: Subsystem A — Baking Quintic Splines & ORCA 3D Avoidance

GNC LEAD: NIKHIL

Key Files: `orca_avoidance.py`, `sutra_fsd_autopilot_node.py` Verified: 120/120 Tests

MATHEMATICAL FORMULATION

- Minimum-snap quintic polynomial splines: $\mathbf{p}(t) = \sum \mathbf{a}_k t^k$ (Jerk < 4.20 m/s³)
- 3D Reciprocal Velocity Obstacle half-planes with static push: $\mathbf{u} = \hat{\mathbf{n}} \cdot \mathbf{v}_{\text{push}} - \mathbf{v}_{\text{rel}}$ (Clearance $\geq 2.80\text{m}$)
- Control Barrier Function (CBF) shield: $\dot{h}(\mathbf{x}) + \gamma h(\mathbf{x}) \geq 0$ enforcing dynamic safety envelope

- Non-Coplanar Cruising Altitudes:** Assigns staggered cruising layers (3.5m, 3.8m, 4.1m, 4.4m, 4.6m) to eliminate 2D collinear stall singularities.
- SutraNeuroFlight:** 0.04ms ONNX network injects dynamic feedforward compensation for 18 m/s wind shear at 50Hz.

```
pytest sutra_ws/src/sutra_gnc/test/ -q # Target: 120 passed in ~4.0s
```

Recipe 2.2: Subsystem B — Brewing Deep JSCC Under -5 dB Jamming

COMMS LEAD: NIKHIL / ROHITH

Key Files: `perceptron_jsc.py`, `mesh_node.py` Verified: 48/48 Tests

THE TECHNICAL MOAT

- Rate-distortion loss: $L = ||\mathbf{x} - \hat{\mathbf{x}}||^2 + \beta R$ (96.9% compression: 512KB $\rightarrow$ 16KB)
- Graceful analog degradation surviving -5 dB RF jamming (PSNR ≥ 41.5 dB) without digital frame freezing.
- SwarmRAFT consensus with 150–300ms heartbeats $\rightarrow$ leader failover < 500ms upon node drop.

- The Defense Answer:** "Digital H.264 freezes completely under 5% packet loss. Deep JSCC maps pixels directly to complex channel symbols—under heavy jamming, video degrades into soft analog grain rather than crashing."

```
python3 scripts/run_deep_jsc_moat_demonstrator.py
```

Recipe 2.3: Subsystem C — Slicing YOLOv8-Nano & WGS84 Raycasting

PERCEPTION LEAD: VEDANTH

Key Files: `detector_node.py`, `bytetrack.py` Verified: 60/60 Tests

RAYCAST FORMULATION

$\mathbf{r}_{\text{world}} = \mathbf{R}_b^w \mathbf{R}_c^b \mathbf{K}^{-1} [\mathbf{u}, \mathbf{v}, 1]^T$ intersecting terrain DEM plane $Z(X,Y) \rightarrow$ WGS84 error < 0.32m at 30m AGL

- ByteTrack Persistence:** 8-state Kalman filters $\mathbf{x} = [\mathbf{u}, \mathbf{v}, \mathbf{s}, \mathbf{r}, \dot{\mathbf{u}}, \dot{\mathbf{v}}, \dot{\mathbf{s}}]^T$ retain survivor identity across occlusion (≤ 1 ID switch across 50 frames).
- Pitch/Roll Correction:** Rotating camera vectors through 6-DOF body attitude matrix $\mathbf{R}_b^w$ prevents 2.5m flat-projection errors.

```
pytest sutra_ws/src/sutra_perception/test/ -q # Target: 60 passed in ~2.4s
```

Recipe 2.4: Subsystem D — Blitting 60 FPS WebGPU Buffers

GCS LEAD: SIVA

Tech Stack: React 18 + Mapbox GL JS + WebGPU Bundle: 226 kB (1.55s build)

React virtual DOM reconciliation cannot handle five 50Hz telemetry streams. SUTRA parses binary WebSocket `ArrayBuffers` directly into `Float32Array` ring buffers, blitting straight to WebGPU canvas draw calls to lock exactly 60.0 FPS.

```
cd sutra_ws/src/sutra_gcs && npm run preview -- --port 3000
```

Recipe 2.5: Subsystem E & F — Plating Pitch & Unit Economics

LEADS: HARIKA & ROHITH

Cost: ₹42,850/drone vs ₹15,00,000 commercial Savings: 97.1% Cost Reduction

Subsystem Component	Hardware Specification	Unit Cost (INR)
Frame & Propulsion	Carbon Fiber 350mm + EMAX 2212 980KV + 20A ESC	₹7,200
Autopilot & Compute	Pixhawk 6C Mini + Companion Edge Board	₹18,400
Dual Vision Sensors	Sony IMX477 (Visual) + FLIR Lepton 3.5 (Thermal)	₹12,500
Mesh Transceiver	802.11s Dual-Band Mesh Module	₹4,750
TOTAL PER AUTONOMOUS SEARCH & RESCUE UAV		₹42,850

Chapter 3: The 3 Evaluation Service Courses (300 Marks)

Evaluation Stage	Time Window	Core Jury Deliverables	Passing Criteria
Course 1: Eval 1 100 MARKS	Day 1 (03-Sep) 05:00 PM onwards	- System Architecture Diagram 232/232 PyTest pass live in terminal - Baseline 5-UAV flight demonstration - SH-DST-05 problem statement mapping	Clean git tree; live test execution; confident architectural defense.
Course 2: Eval 2 100 MARKS	Day 2 (04-Sep) 02:00 PM onwards	100% Closure of Eval 1 Jury feedback - Disturbance rejection (18 m/s wind) -5 dB RF jamming video test - Single motor failure fallback demo	JURY_FEEDBACK_TRACKER.md showing tagged commit hashes for all feedback.
Course 3: Eval 3 100 MARKS	Day 3 (05-Sep) 08:30 – 11:00 AM	- Live 5-UAV Ring Crossing Demo - Sub-0.32m WGS84 Geolocation Raycast - Unit Economics BOM defense - Final Stage Master Pitch (Harika)	Zero collision ring crossing; locked 60 FPS HUD; flawless jury Q&A defense.

Chapter 4: Emergency Fire Suppression (Worst-Case Fallbacks)

Disaster Scenario 1: Venue Wi-Fi Drops Completely

HIGH PROBABILITY

- Immediate Action: Do not waste time debugging campus network.
- Offline Activation: Run: `python3 -m http.server 8000 --directory /home/nikhil/Desktop/Project\ SUTRA`.
- Open Offline Portal: Open `http://localhost:8000/SUTRA_OFFLINE_PORTAL.html` in Chrome. It runs WebGPU simulations, rate-distortion curves, and 5-UAV ring crossing entirely in browser memory without network.

Disaster Scenario 2: Jury Demands: "Kill Drone 3 Right Now"

ADVERSARIAL TEST

- Trigger Failure: Nikhil executes: `ros2 topic pub --once /sutra/uav_3/motor_failure std_msgs/msg/Bool "{data: true}"`.
- Autonomous Recovery: `motor_failure_fallback_node.py` detects zero RPM within 80ms, commands UAV 3 into controlled spin-damped descent, while remaining 4 UAVs dynamically recalculate search polygons to close the gap.

The 15-Second Master Verification Gate

PRE-DEMO VERIFICATION

```
# 1. Run Complete 232-Test Deterministic Suite:
pytest sutra_ws/src/sutra_*/test/ -q

# 2. Verify Zero Uncommitted Code (Cardinal Law):
git status

# 3. Check GCS Production Build:
cd sutra_ws/src/sutra_gcs && npm run build
```

Library Workstation 4-Screen Hardware Layout

STATION SETUP

Machine	Teammate	Active Screen Display	Defense Ownership
ASUS TUF A15 (RTX 3050)	 Nikhil	Gazebo Sim 8 Digital Twin / Live 5-UAV ROS 2 Node Graph	GNC Math, ORCA 3D, PX4 Offboard & Deep JSCC
HP Victus (RTX 4050)	 Rohith Kumar	Deep JSCC Moat Benchmark & TensorRT Inference Monitor	GPU Runner & Desk Anchor (Rule 3.4)
MacBook Pro (Apple Silicon)	 Harika	14-Slide Master Pitch Deck + JURY_FEEDBACK_TRACKER.md	Pitch Delivery, Feedback Logging, Unit Economics
Lenovo Laptop (Intel i5)	 Siva /  Vedanth	SUTRA 3D GIS GCS HUD (Port 3000) & WGS84 Target Stream	GCS Mapbox Interface & YOLOv8-Nano Detections

30-Second Winning Pitch Script (Harika/Nikhil): "Respected Jury, Project SUTRA solves the three fatal failure modes of modern search-and-rescue: single-drone coverage bottlenecks, GPS loss in canyons, and video stream collapse under RF jamming. Powered by ORCA 3D reciprocal collision avoidance with non-coplanar cruising layers, analog Deep Joint Source-Channel Coding, and terrain-corrected WGS84 raycasting, SUTRA provides fully decentralized, collaborative search capability at ₹42,850 per drone—97% cheaper than enterprise hardware. Every single claim is backed by 232 automated tests running in 15 seconds."